



BobSim

TransientEval Lateral Transient Response

Step response and sustained-sine FRF

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Run Notes

- Representative step steer response
- Continuous sinusoidal responses
- Rise time uses 50 percent input to 90 percent response thresholds
- Frequency delays are reported at 0.5 Hz and 1.0 Hz
- Gain and phase slopes summarize the FRF trend
- Steering shown in degrees for readability

TransientEval Metrics Summary 15.0 m/s
Time Domain - Step Response

STEP RESPONSE			STEADY-STATE / OVERSHOOT		
a_y Rise Time (50-90%)	0.07	s	a_y Peak	3.41	m/s ²
Sideslip Rise Time (50-90%)	0.18	s	a_y Steady-State	2.83	m/s ²
Yaw Rise Time (50-90%)	0.06	s	a_y Overshoot	20.5	%
Yaw Peak Response Time	0.10	s	Yaw Steady-State	0.190	rad/s
a_y DC Gain	33.57	$\frac{\text{m/s}^2}{\text{rad}}$	Yaw Overshoot	0.035	rad/s
Sideslip DC Gain	0.032	rad/rad	Roll Steady-State	0.004	rad
Yaw DC Gain	2.22	(rad/s)/rad	Roll Overshoot	0.001	rad
Roll DC Gain	0.044	rad/rad	Settling Time	1.31	s

TransientEval Metrics Summary 15.0 m/s

Frequency Domain - Sustained Sine

FREQUENCY RESPONSE - CORE

DC Gain (a_y)	33.57	$\frac{\text{m/s}^2}{\text{rad}}$
DC Gain (r)	2.22	(rad/s)/rad
Peak Gain (a_y)	34.70	$\frac{\text{m/s}^2}{\text{rad}}$
Peak Freq (a_y)	1.00	Hz
Bandwidth (-3 dB)	1.00	Hz
Gain Slope (a_y)	0.94	dB/dec
Gain Slope (r)	0.31	dB/dec
Phase Slope (a_y)	10.56	deg/dec
Phase Slope (r)	16.53	deg/dec

DELAY / COUPLING

Ay/Yaw Delay @ 0.5 Hz	0.014	s
Yaw/Steer Delay @ 0.5 Hz	0.027	s
Difference @ 0.5 Hz	-0.013	s
Ay/Yaw Delay @ 1.0 Hz	0.016	s
Yaw/Steer Delay @ 1.0 Hz	0.027	s
Difference @ 1.0 Hz	-0.012	s
Gain Variation	3.3	%
Fit Error (a_y)	2.58e-03	-
Fit Error (r)	9.00e-04	-

TransientEval Metrics Summary 20.0 m/s

Time Domain - Step Response

STEP RESPONSE

a_y Rise Time (50-90%)	0.10	s
Sideslip Rise Time (50-90%)	0.09	s
Yaw Rise Time (50-90%)	0.06	s
Yaw Peak Response Time	0.11	s
a_y DC Gain	53.57	$\frac{\text{m/s}^2}{\text{rad}}$
Sideslip DC Gain	-0.016	rad/rad
Yaw DC Gain	2.68	(rad/s)/rad
Roll DC Gain	0.070	rad/rad

STEADY-STATE / OVERSHOOT

a_y Peak	4.95	m/s^2
a_y Steady-State	4.52	m/s^2
a_y Overshoot	9.5	%
Yaw Steady-State	0.227	rad/s
Yaw Overshoot	0.030	rad/s
Roll Steady-State	0.006	rad
Roll Overshoot	0.001	rad
Settling Time	1.22	s

TransientEval Metrics Summary 20.0 m/s

Frequency Domain - Sustained Sine

FREQUENCY RESPONSE - CORE

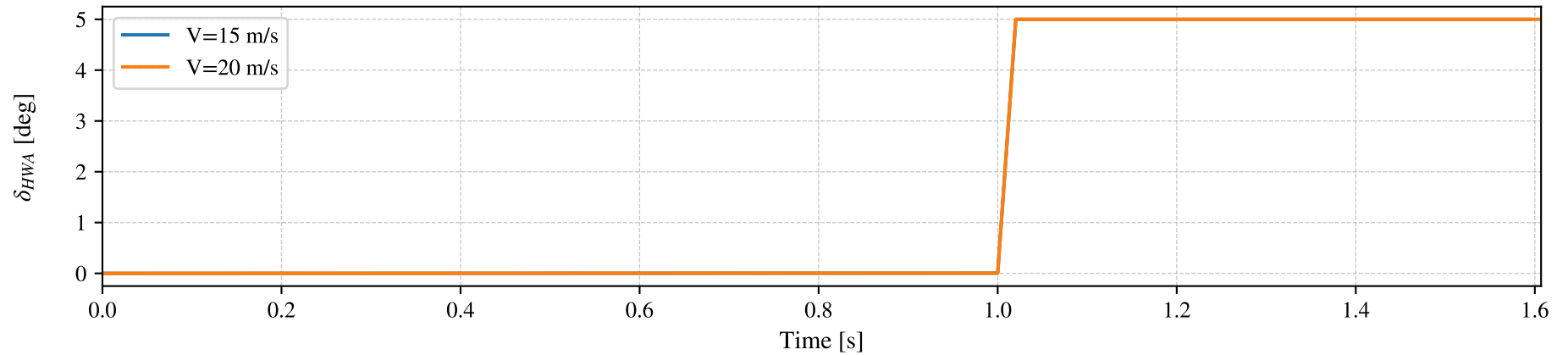
DC Gain (a_y)	53.57	$\frac{\text{m/s}^2}{\text{rad}}$
DC Gain (r)	2.68	(rad/s)/rad
Peak Gain (a_y)	53.61	$\frac{\text{m/s}^2}{\text{rad}}$
Peak Freq (a_y)	0.75	Hz
Bandwidth (-3 dB)	1.00	Hz
Gain Slope (a_y)	0.01	dB/dec
Gain Slope (r)	0.13	dB/dec
Phase Slope (a_y)	24.36	deg/dec
Phase Slope (r)	18.62	deg/dec

DELAY / COUPLING

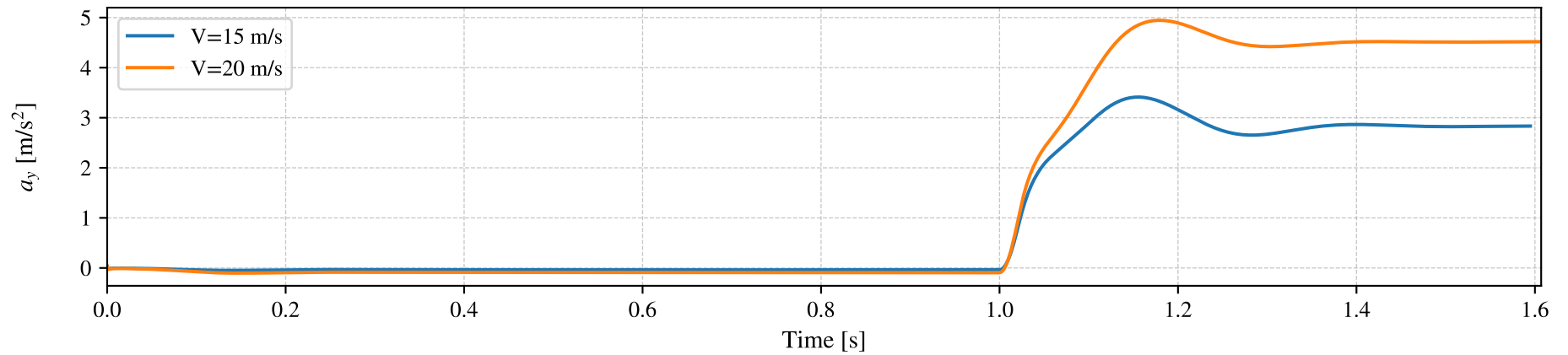
Ay/Yaw Delay @ 0.5 Hz	0.039	s
Yaw/Steer Delay @ 0.5 Hz	0.031	s
Difference @ 0.5 Hz	0.008	s
Ay/Yaw Delay @ 1.0 Hz	0.040	s
Yaw/Steer Delay @ 1.0 Hz	0.031	s
Difference @ 1.0 Hz	0.009	s
Gain Variation	0.1	%
Fit Error (a_y)	5.97e-03	-
Fit Error (r)	3.59e-03	-

Step Steer Response (TransientEval)

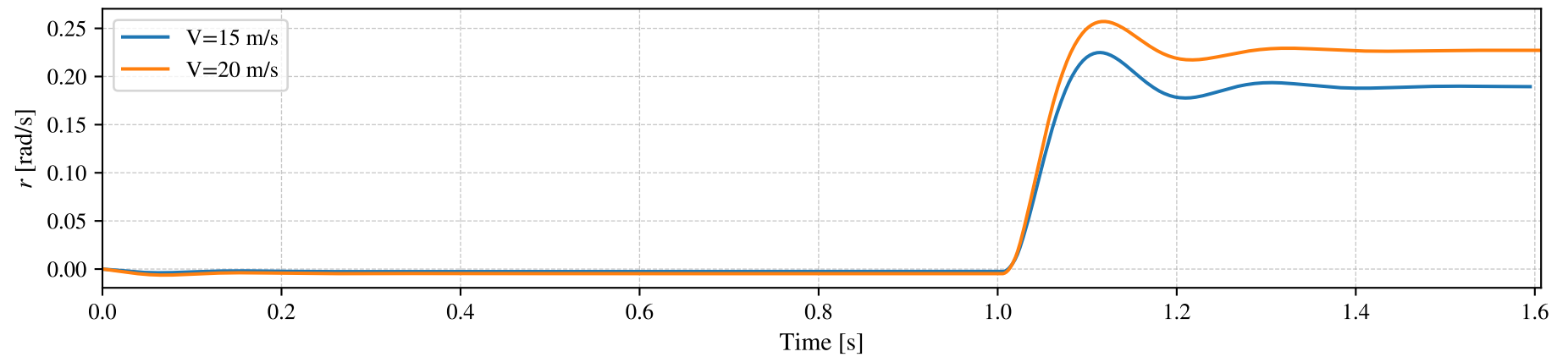
Steering Wheel Angle δ_H



Lateral Acceleration a_y

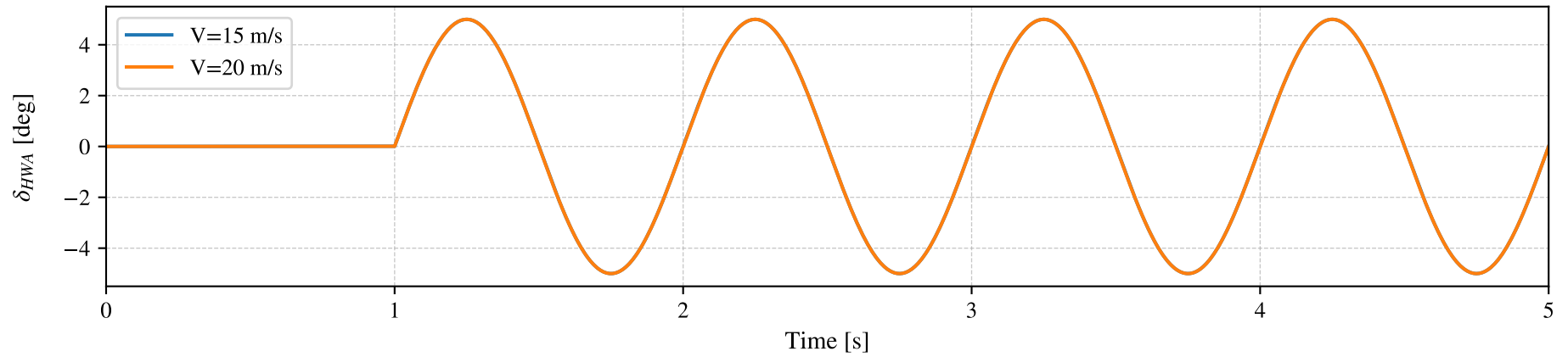


Yaw Velocity r

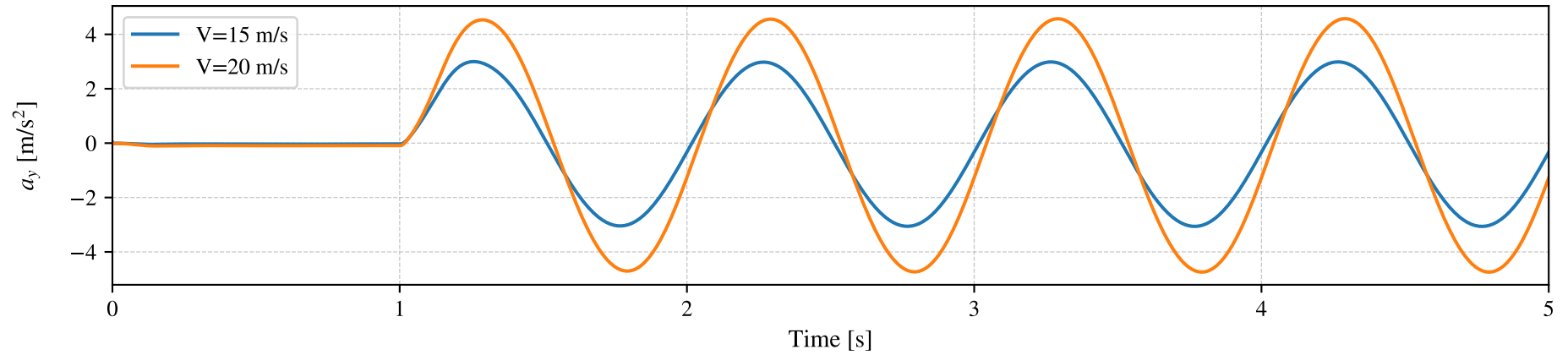


Continuous Sinusoidal Response

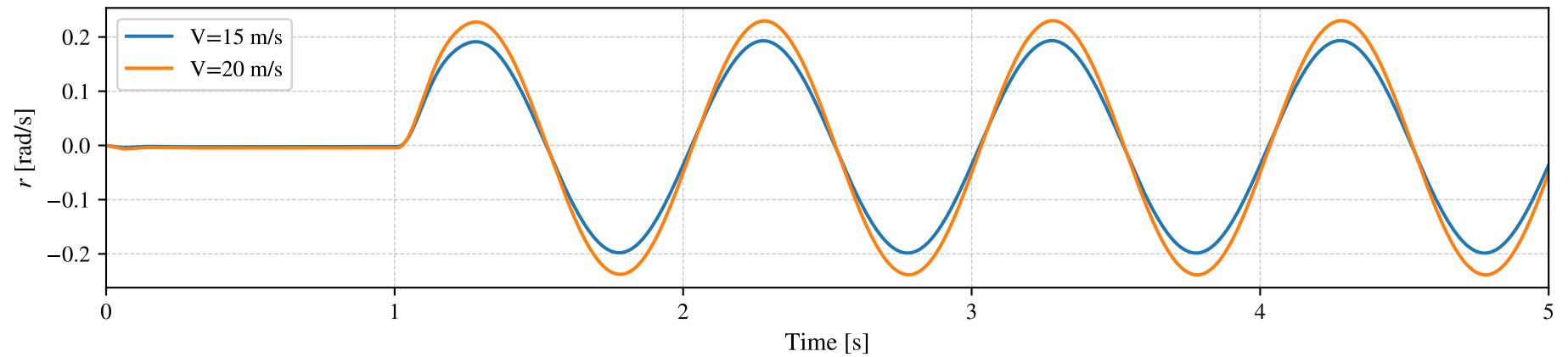
Steering Wheel Angle δ_H



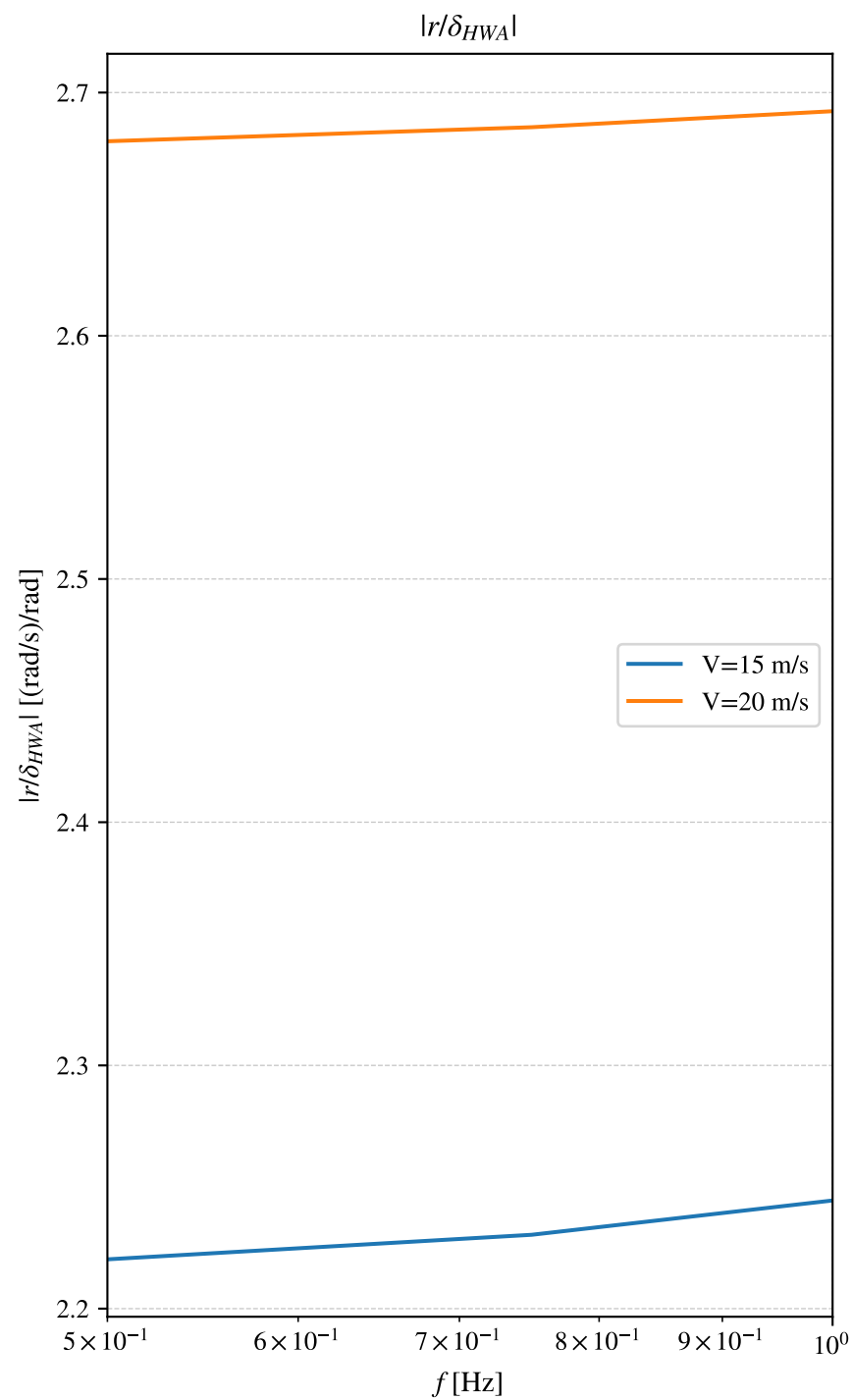
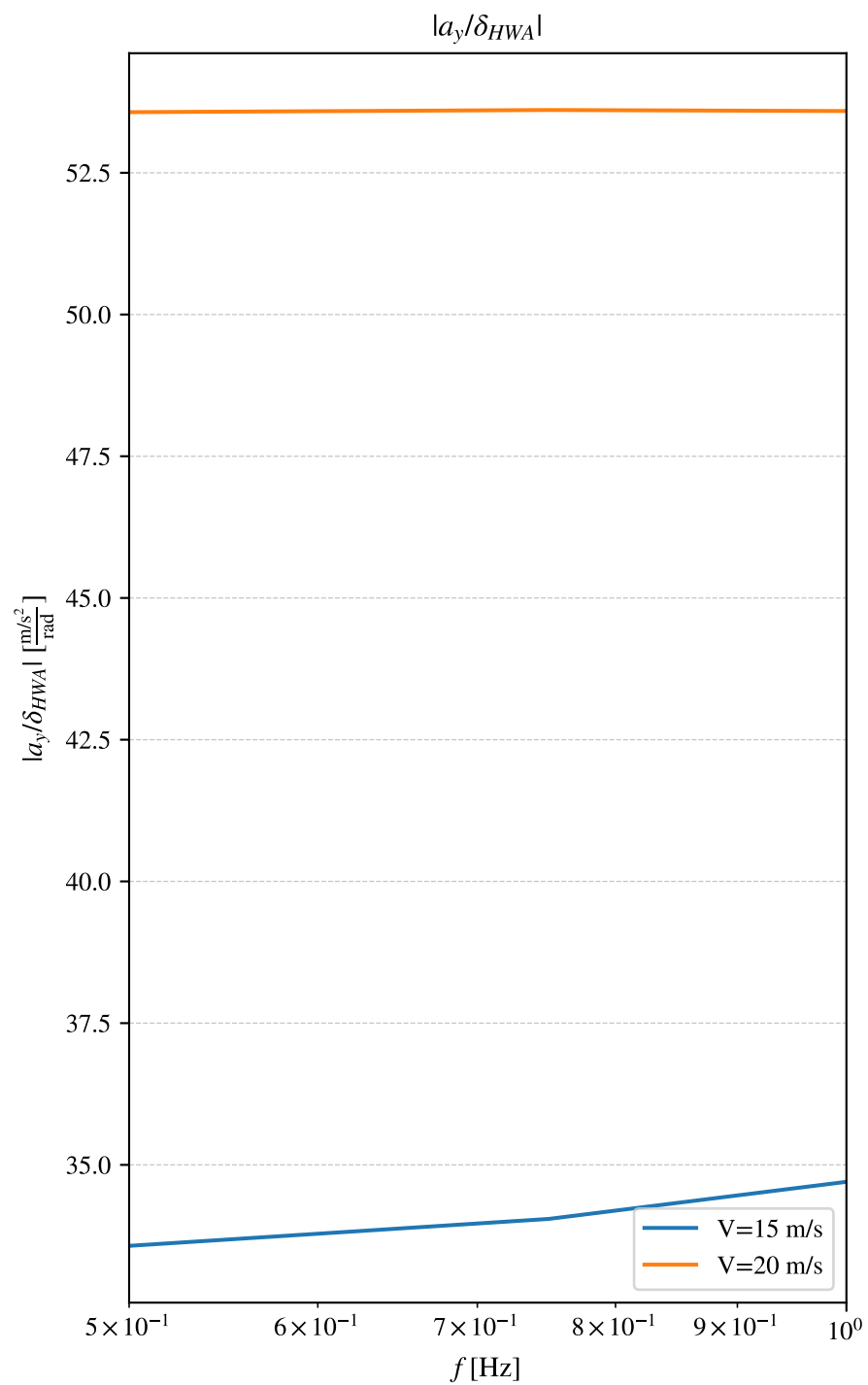
Lateral Acceleration a_y



Yaw Velocity r

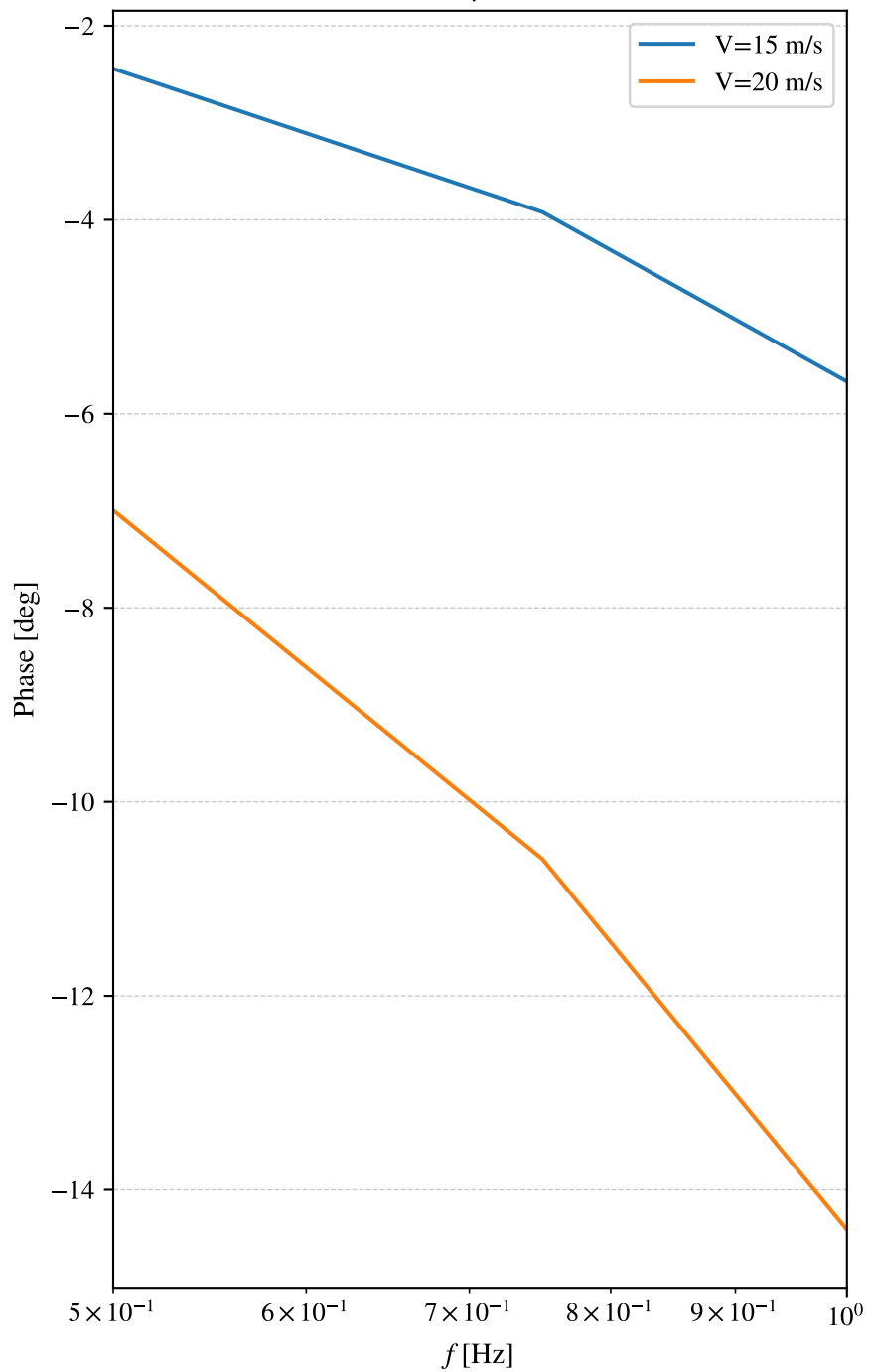


Frequency Response — Gain (TransientEval)



Frequency Response — Phase (TransientEval)

Phase: a_y vs δ_{HWA}



Phase: r vs δ_{HWA}

